



An assisted propulsion device of vessel utilizing wind energy based on Magnus effect

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ABSTRACT

This research proposes a new type of vessel assisted propulsion device, which is a polygonal sail composed of sails and cylinders, and the sail is essentially a super large Flettner rotor that can be deformed. The main structure of the device is modelled using SolidWorks software and a three-dimensional computational domain containing the polygonal sails is established to analyse the beneficial effects that the polygonal sails provide to the vessel. The polygonal sail is simulated and calculated using CFD commercial code under various working conditions. A comparative analysis of the lift coefficients and drag coefficients for nine groups shapes of polygonal sails, which found that the greater number of the sail sides, the greater the lift coefficient. After a comprehensive comparison, it is concluded that the 16-sided sail has better practical application value for a 300000-ton tanker. Further numerical simulations are operated for the model of 16-sided sail, and the maximum lift that the sail can provide to the vessel is 590 kN. When the spin ratio k is 1, 8-level wind conditions, the maximum propulsive power of the sail is 2005 kW. In addition, the Reynolds number is also one of the main factors affecting the force coefficient of a polygonal sail. As the Reynolds number increases, the lift-to-drag ratio of the sail becomes smaller.

1. Introduction

With the increasing number of international merchant vessels, the burning of heavy fuel oil by vessels is causing increasing air pollution. Therefore, in recent years, the International Maritime Organization (IMO) has put forward higher requirements on the emissions of vessels [1–3]. It is imperative to reduce the emissions of vessels, and find new and clean alternative energy sources [4, 5].

At present, the clean energy actually used on merchant vessels is mainly wind energy and solar energy. On the one hand, regarding the utilization of solar energy on vessels, the main method is to install solar panels on the deck to realize the recovery and use of solar energy, so as to provide electricity for vessels [6–9]. On the other hand, as renewable energy, wind energy has the advantages of large reserves, wide distribution, and zero pollution [10, 11]; besides, because the ocean is relatively open and the resistance of wind is smaller, wind energy resources are more abundant at ocean than on the land; vessel underway has a unique advantage in applications of wind energy resources [12–14].

Some shipowners had installed devices to capture wind energy on navigating merchant vessels to convert wind energy into propulsion

contribution for vessel navigation, achieving savings in fuel. The more mature devices for capturing wind energy mainly include: traditional sails, Flettner rotors, kite sails and airfoil sails [15–18]. Among them, the Flettner rotor is a very effective way to capture wind energy. It was used on vessels as early as 100 years ago. The Flettner rotor is generally installed vertically on the vessel deck. It relies on the motor to drive the rotor to rotate, then, due to the Magnus effect, a heel force perpendicular to the direction of the airflow is generated as the wind blows across them. The magnitude and direction of the Flettner rotor's force can be adjusted to provide thrust for the vessel by changing the speed and direction of rotation of the rotor. This is principle of rotor works.

In the past few years, a larger number of merchant vessels have installed Flettner rotors. In January 2018, the 6400-ton bulk carrier “Afros” customized by the Greek shipowner Victoria Steamvessel was equipped with 4 Flettner rotors with the rotor height of 18 meters and a maximum speed of 450 rpm. According to the data collected by the wind speed and wind direction sensors on the vessel to adjust the speed and steering of the rotor accordingly to obtain the maximum thrust. In theory, it can save 4-ton main engine fuel consumption per day at the same speed. In April of the same year, a “Viking Grace” of Viking Cruises was installed with a 24 m high and 4 m diameter rotor sail, which

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Nomenclature			
D	Diameter of device	F_T	Thrust
H	Height of device	C_L	Lift coefficient
A	Vertical cross section area of sail	F_L	Lift
ρ	Fluid density	C_D	Drag coefficient
ν	Kinematic viscosity is 1.45×10^{-5} Pa/s	F_D	Drag
V_{ve}	Vessel speed	P_L	Output power
V	Wind speed	C_M	Moment coefficient
$Re = \frac{V \cdot D}{\nu}$	Reynolds number	P_{motor}	Consumed power of the motor
U_{tan}	Tangential velocity of sail	$P_{prop.}$	Propulsive power
$k = \frac{U_{tan}}{V}$	Spin ratio	p	Pressure
y^+	A nondimensional wall distance	p_∞	Free flow pressure
		$C_p = \frac{p - p_\infty}{0.5 \rho V^2}$	Pressure coefficient
		θ	Wind angle

become the world's first passenger vessel to adopt rotor sail technology. In August 2018, Maersk announced the installation of two rotor sails with a height of 30 m and a diameter of 5 m on one of its L2 tankers. This is also the largest rotor sail so far. It can be seen from the above cases that the Flettner rotor has a good effect on assisting navigation [19, 20].

The rotor sail uses the force generated by the Magnus effect to assist the vessel on voyage. Analysis of the factors influencing the Magnus effect, it is not difficult to find that the tangential speed of the rotor, the environmental wind speed and the size of the rotor can directly affect its propulsion contribution. If the area of the rotor sail in contact with the wind is increased, it will effectively increase the thrust generated by the vessel. Moreover, the width of the general Flettner rotor currently installed and applied on the vessel is about 1.5–5.5 m, which accounts for about 1/10 of the deck width, the size of rotor can still be expanded. However, if you directly increase the size of the general Flettner rotor, although it can increase the propulsion effect, it will disturb the work of the vessel's deck machinery and the driver's field of vision. What's more, when the wind of bow direction appears, the larger-size rotor sails will produce resistance. Therefore, considering these above factors, the size of a traditional Flettner rotor is generally not very large.

In order to solve the above problems, this research proposes a new assisted propulsion device of the vessel using the Magnus effect. It is a super large Flettner rotor composed of rotating cylinders and a sail. It has a larger projection area and can theoretically produce a better propulsion effect. In this study, the device will be modeled, and aerodynamic simulation will be performed on it through commercial software Fluent [21, 22].

Many scholars have carried out experiments and related simulations on the Flettner rotor and Magnus effect. In reference [23], Copuroglu and Pesman presented the effect of the Flettner rotor on the lateral motion of the vessel and the effect of the lateral motion on the performance of the Flettner rotor in the beam. Their results show that the lifting power performance of the Flettner rotor decreases with increasing heeling angle. Tillig et al. [24] analyzed 11 different Flettner rotors in tankers and roll-on/roll-off ship. The study had one result that 30% fuel savings for tanker, 14% fuel savings for ro-ro vessel, which is a more desirable result. For the study of the Flettner rotor using Self-propulsion of an object, the optimal control problem for several types of inputs was solved using a genetic algorithm by Ramodanov et al. [25]. Perkovic et al. [26] proposed that a wind turbine using the Magnus effect, and the corresponding 2D pneumatic and process dynamics were analyzed for each airborne module (In reference [26], the module essentially is a two-dimension circle). The results of the study are that wind turbine is feasible for power production. Pezzotti et al. [27] found that the results of an experiment to study the flow behavior around different cylindrical bodies with 4, 6, 8, and 10 sides. Magnus effect of non-cylinder is studied, and explaining fluid dynamics behavior through flow visualization tests. Similarly, polygonal circular sails are discussed in our

paper. Correa et al. [28] proposed Magnus wind turbine emulator, the special case of the horizontal axis wind turbine and the main physical principles involving the Magnus phenomenon are discussed. The project has helped to encourage further research on wind turbines using the Magnus effect. In reference [29], a preliminary assessment of numerical setup has been conducted, spin ratio, aspect ratio and tip geometry is considered to be the main experimental variable, and it provides guidance for further research.

2. Structure model of device

A new type of assisted propulsion device for the vessel is proposed, which is composed of rotating cylinders and sails together, the device model is built using SolidWorks software as show in the Fig. 1. (The figure shows the schematic diagram of the device with eight cylinders, the number of rotating cylinders in the device can be optimized and adjusted according to the actual situation). In the following, the main structure and principle of the device will be briefly described, taking the device with eight rotating cylinders as an example. The device is mainly composed of multiple rotating cylinders, sails, platforms, tracks, motors and hydraulic motors. Among them, the sails surround the eight rotating cylinders, and the sails and the rotating cylinders cooperate with each other (the connection between sail and cylinder is similar to the tensioning connection in a belt drive equipment); when the multiple cylinders are driven by the motors to rotate together, the sails also undergo rotational motion, thus the Magnus effect is produced on the device. Lastly, the main equipment is mounted on the platform, such as multiple rotating cylinders, sails, tracks and so on.

The position of the rotating cylinders and the shape of the sails in the device can be changed depending on the actual wind conditions. The working principle of the device is briefly described as follows: each rotating cylinder is set in the track using the slot, and the cylinder can slide along the track under the drive of the hydraulic motor, so that different shapes of sails can be constructed. When the wind comes from the bow direction of the vessel, the state of the device is upwind condition, as in Fig. 1(c), the sail has the smallest projected area along the wind direction, and the it is not rotating at this time. When the wind comes from the port and starboard side of the vessel, or the wind comes from the bow, the device becomes to an approximate circle, and it is regarded as a polygon condition, as in Fig. 1(a). Besides, the rotation speed of the sail is judged according to the wind sensor in the same way as the traditional Flettner rotor, so it is not repeated here.

The proposed assisted propulsion device using the Magnus effect has the following advantages. Firstly, the diameter of the outer tangent circle of the sail is 50 m, and height of sail is 14 m, which solve the technical problem that the diameter of the traditional Flettner rotor is small and the wind area cannot be too large, so the device can generate a large Magnus force when the vessel is sailing with the wind on the port

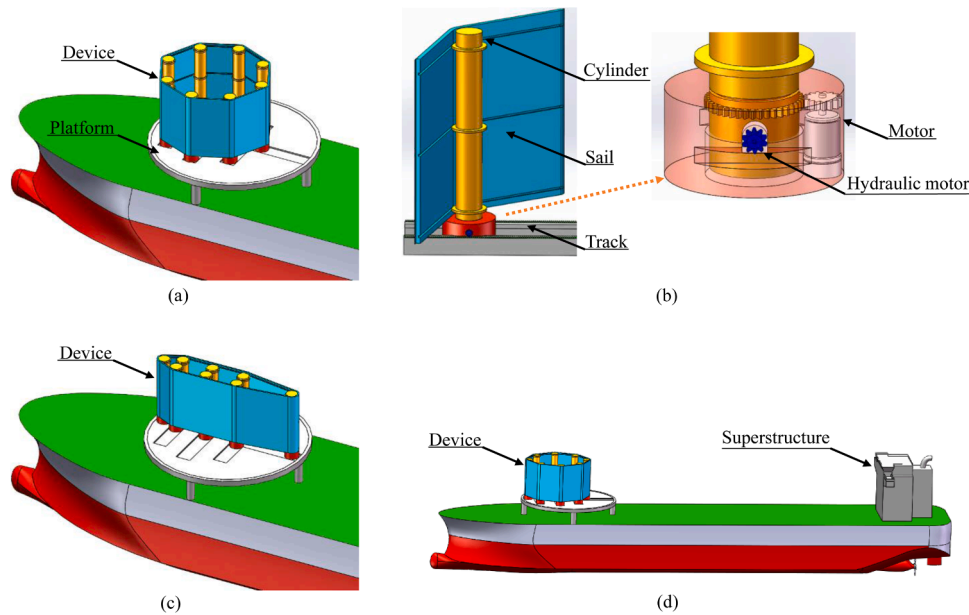


Fig. 1. Assisted propulsion device: (a) Polygon condition, (b) Single rotating cylinder, (c) Upwind condition, (d) Device's position on the vessel.

side or the starboard side, which provides part of the power for the vessel's sailing, reduces the fuel consumption, and realizes the energy saving. Secondly, according to different wind directions, the shape of the sail can be changed, and when the vessel is sailing into the bow wind, the large resistance problem of the general Flettner rotor is solved by reducing the projected area of the upwind side of the sail. Thirdly, the sail is supported by multiple cylinders, thus a larger size Flettner rotor is constructed (The sail is made of light and flexible material). Under the same environmental conditions, the device generates more force compared to a traditional Flettner rotor of equal mass.

In this study, the 300000-ton oil tanker "Xin Hui Yang" is used as the base vessel, and the main parameters of this vessel are shown in Table 1. In order to better harvest the wind energy, the width of the platform is set as close as possible to the width of the vessel; the diameter of the outer tangent circle of the device is 50 m; the height of sail is 14 m. It is worth noting that the sail in the device shown in Fig. 1 is octagonal; the polygonal sails are discussed below keeping the diameter of the tangent circle unchanged at 50 m and the height of the device unchanged at 14 m. In addition, all polygonal sails described in the paper have a height of 14 m.

3. Numerical calculation model

3.1. Physical model

The device in this study has two shapes when the vessel faces different wind conditions, namely, upwind condition and polygon condition. The polygonal device is capable of generating lift to contribute to sail, so the following simulations focus on the response of the sails in the polygonal device when subjected to sidewind perpendicular to the vessel's hull. The Fig. 2 shows the vessel force diagram and wind direction diagram in sidewind. In this study, the more common wind level of 4-8 is taken as cases set in the simulation calculation, and the wind

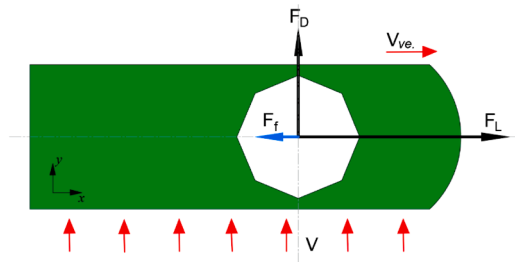


Fig. 2. Vessel force and wind direction diagram.

speed corresponding to the specific wind level is shown in Table 2.

The power consumed by the motor that rotates the sails in the device and the force acting on the sails together determines the value of the power that the device can generate. Considering only the vertical side-wind acting on the hull, the lift generated by the device is along the direction of sailing (as in Fig. 2), and the positive thrust that the vessel receives from the device is the Magnus lift (F_L), so the lift directly affects the output power of the device (the power transferred by the sail in the ideal case). The lift (1) and drag (2), as well as the output power of the device (3) and the power consumed by the motor that drives the sail rotation (4), are given by the following equations [30].

$$F_L = \frac{1}{2} * \rho * A * V^2 * C_L \quad (1)$$

$$F_D = \frac{1}{2} * \rho * A * V^2 * C_D \quad (2)$$

$$P_L = F_L * V_{ve} \quad (3)$$

$$P_{motor} = 0.5 * \rho * A * V^3 * C_M * k \quad (4)$$

Propulsive power ($P_{prop.}$) is calculated as the difference between the

Table 1
Parameters of the tanker "Xin Hui Yang".

Parameters	value	Parameters	value
length overall (m)	330	Draft (m)	21.5
Breadth moulded (m)	60	Operating speed (kn)	12
Deadweight (ton)	307664	Main engine power (kW)	25480

Table 2
Parameters of wind speed.

Wind level	4	5	6	7	8
Wind speed (m/s)	7	10	13	16	20

output of the device and the power consumed by the rotation of the sails, and its equation is as follow:

$$P_{prop.} = P_L - P_{motor} \quad (5)$$

3.2. Computational domain and boundary conditions

In this experiment, CFD software Fluent 18.0 is applied as a simulation platform, and the lift coefficient, drag coefficient and propulsive power of the device are calculated [31, 32]. To be specific, a three-dimensional computational domain containing polygons is created and the SIMPLE algorithm is used to calculate the N-S equation to obtain more reliable results. To ensure the accuracy of the simulation, a second-order discretization scheme is utilized; the turbulence model is chosen as the more commonly used Realizable $k-\varepsilon$ model. In this paper, the RANS equation is used as the basic equation to solve the domain around the polygonal sail, and the continuity Eq. (6) and momentum Eq. (7) are used [33, 34].

$$\nabla \cdot U = 0 \quad (6)$$

$$\rho \bar{u}_i \frac{\partial \bar{u}_i}{\partial x_j} = \rho \bar{f}_i + \frac{\partial}{\partial x_j} \left[-\bar{p} \delta_{ij} + \mu \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \rho \overline{u'_i u'_j} \right] \quad (7)$$

where U is the relative velocity; ρ is the density of fluid; μ is liquid viscosity; $\bar{f}_i$ is mass force term; $\bar{p}$ is static pressure; δ_{ij} is Kronecker delta; $-\rho \overline{u'_i u'_j}$ is apparent stress owing to the fluctuating velocity field, generally referred to as the Reynolds stress.

As shown in Fig. 3, the whole computational domain is an approximate cylindrical shape, which has a diameter of $16D$. D is the diameter of the outer tangent circle of the sail; the height (thickness) of the cylindrical area is equal to the height of the device of $14m$. A Hexa grid is used for the whole area, and a fine area is set in the middle of the computational domain. The sail is surrounded by a circular rotating area of $1.5D$ diameter, this area represents the wind field generated by the sail, which is also adopted by Marco et al., which is particularly suitable for rotational motion [35]. In order to ensure the accuracy and smoothness of the calculation, five boundary layers are set up around the polygon sail, as in Fig. 3(b). In addition, the nondimensional wall distance (y^+) approximately equal to 1; the Max-Skewness is 0.45; the blocking ratio is approximately equal to 6.25%.

As in Fig. 3(a), the bottom of the computational domain is the inlet

boundary, which is the velocity-inlet and its direction is positive to the y -axis (It is a sidewind perpendicular to the hull). The top of the computational domain is the pressure-outlet, which has a pressure value of one atmosphere. The sail belongs to no-slip wall. The left and right sides of the computational domain contain $2.5D$ long walls, they are no-slip wall, these walls are the dividing line between the inlet and outlet. In order to ensure good convergence and improve computational efficiency, the base of the Flettner rotor and an upper boundary above the rotor are no-slip wall. The rotation area (red area in Fig. 3(a)) is rotated counterclockwise around the z -axis.

3.3. Grid independence analysis

The grid independence study was set up for four sets of comparison cases with 600000, 700000, 800000 and 1000000 cells. Under the condition of 6-level wind and spin ratio $k = 2$, the pressure coefficients (C_p) of the octagonal sails were used as the scale of comparison grid independence study. Fig. 4(a) shows the pressure coefficient curves corresponding to each of the four groups of grids. The results of the grid independence study are shown in Table 3. Through comparison and analysis, we find that after the grid number reaches 800000, further increasing the grid number basically has no effect on C_p ; comparing the example of grid number 1000000, the error of its value is less than 0.3%. Therefore, within the reasonable range of the calculation accuracy and efficiency are considered, the number of grids of about 800000 is chosen for the subsequent simulation.

In general, the time step simulation time is 30 s for each case (Time Step Size: 0.1 s (20 iterations/step); Number of Time Steps: 300). The required run time of the computer (with Intel Core i7) is about 6-9 h for one case.

Due to the objective conditions limiting the experimental possibility, setup chosen of CFD is validated by using benchmark experimental data on Flettner Rotors. An example of an experiment that we applied CFD to mimic was done by Bordogna et al. [36]. Specifically, the experimental conditions for the rotating cylinder are $H = 3.73m$, $D = 1m$ and $Re = 1.0 \times 10^6$, k is 0, 0.5, 1, 1.5, 2 respectively. The simulation results are shown in Fig. 4(b) "CFD results vs experimental data", and the percentage error between Exp. and CFD in Table 4, it can be seen that both results are generally consistent. Similarly, the polygon sail simulation setup is consistent with the CFD experimental setup of the rotating cylinder. Therefore, the study's setup chosen is also somewhat accurate.

In addition, a comparative analysis was made as to whether the

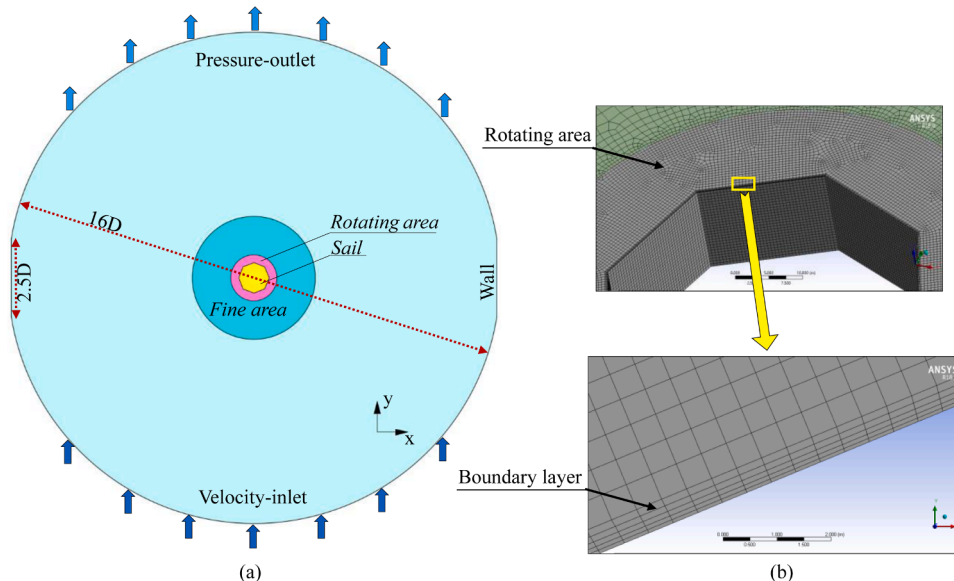


Fig. 3. Computational domain: (a) Boundary conditions and (b) Boundary layer.

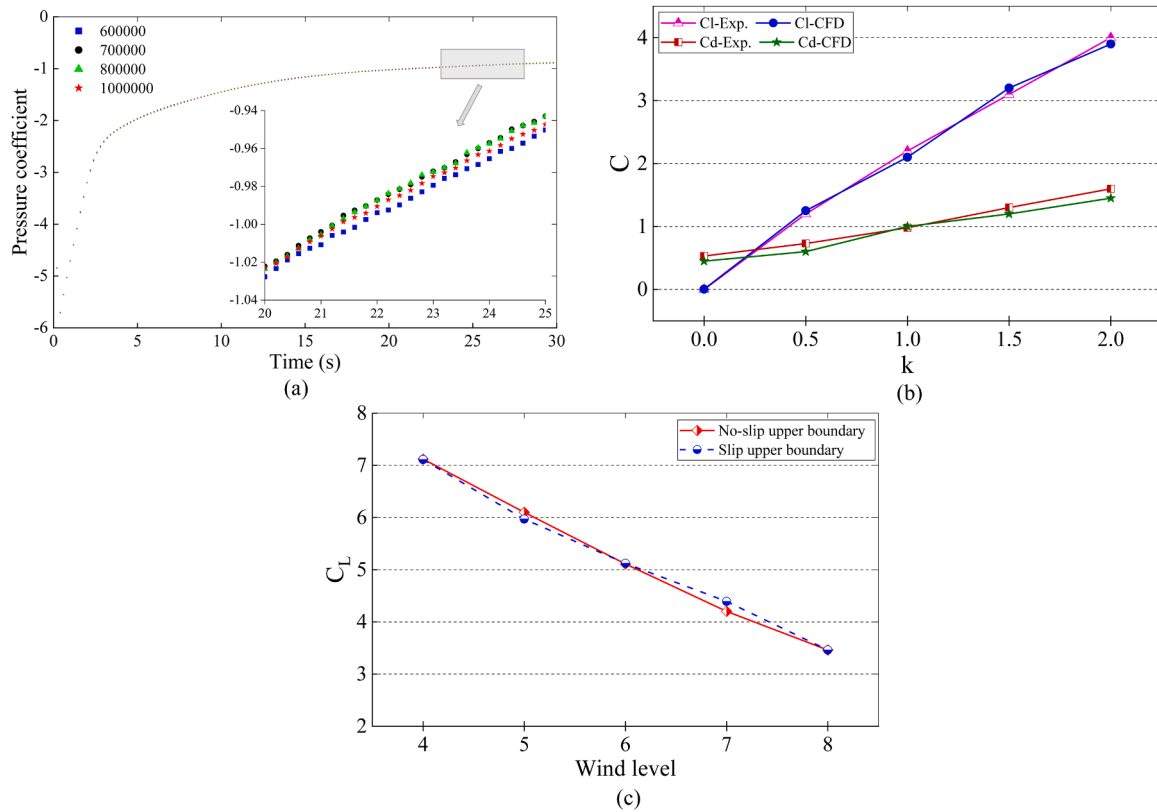


Fig. 4. Grid independence analysis: (a) the pressure coefficient curves corresponding to each of the four groups of grids; (b) CFD results vs experimental data; (c) the comparison of upper boundary condition for the computational domain.

Table 3

The result of the grid convergence study.

Scale	Cells number	Average- C_p (20–25 s)
Coarse	600000	-0.988
Medium	700000	-0.979
Fine	800000	-0.981
Very fine	1000000	-0.984

Table 4

The error between Exp. and CFD ($|Exp.-CFD|/Exp.$).

k	0	0.5	1.0	1.5	2.0
C_L	0.5%	4.2%	4.5%	3.2%	2.5%
C_D	15.1%	17.0%	2.0%	7.7%	9.4%

upper boundary of the computational domain in the study should be applied to the slip or the no-slip boundary condition. For the computational domain model, slip upper boundary and no-slip upper boundary produce two results that are same in terms of lift coefficients, as shown in Fig. 4(c) (where the sails are 16-sided, the wind is pointing vertically on one side of the vessel, the spin ratio $k = 2$, and the wind levels are class 4–8). As a result, in order to obtain a more closed computational domain and better computational efficiency, the no-slip conditions are used for both the upper and base boundaries of the Flettner rotor.

4. Analysis of simulation results

In the following, the aerodynamic characteristics of nine groups of different polygonal sails are explored, on the one hand, in order to further compare and analyze the assisted propulsion contribution produced by the polygonal sails, the Magnus effect of the polygons is

researched. On the other hand, the device itself has consisted of a lightweight sail and vertical cylinders, the more the number of cylinders (more the number of sides of the polygon), the greater the mass and construction cost of the device itself, and the more inconvenient of the sail deformation process. Therefore, the following nine groups of polygonal sails were compared and analyzed, as in Table 5, with the aim of finding a better solution that not only ensures the sails produce as much assisted propulsion contribution as possible, but makes the number of cylinders fewer.

4.1. The performance of the device under three typical wind direction conditions

When a vessel sails in the ocean, it will be affected by wind in different directions in different environment. When the device faces the incoming wind in different directions, it can change the shape of the sail by changing the position of the cylinder, so as to achieve better use of wind energy (sidewind condition) or the purpose of decreasing resistance (upwind condition). Here is a simple exploration of the sails in the device under three typical conditions of the vessel's upwind, sidewind and downwind. When the vessel is upwind condition, the device is shown in Fig. 1(c); when the vessel is sailing in sidewind and downwind conditions, the shape of device is shown in Fig. 1(a). In addition, the sail produces a rotating motion only in the case of sidewind, and the spin ratio k is 2 here. The lift coefficient, lift and pressure distribution of the sail under the three wind directions are given, as shown in Figs. 5 and 6. Among them, in order to form a better comparative analysis, the three wind speeds used here are all 13 m/s; here an octagonal sail is used.

Fig. 5(a) and (b) are the lift coefficients and lift of the sails in the three conditions, respectively. When the vessel is in an upwind condition, the lift coefficient of its sail is -3.2, and its lift is -2 kN, which means that the device produces resistance. In the upwind condition, the resistance of the sail is small, which is 2 kN. When the vessel sails is

Table 5
The nine groups of polygonal sails.

Type	Number of sides	Cross sections	Type	Number of sides	Cross sections
Hexagonal Prism	6		16-Sided Prism	16	
Octagonal Prism	8		18-Sided Prism	18	
Decagonal Prism	10		20-Sided Prism	20	
12-Sided Prism	12		Circular Cylinder	n	
14-Sided Prism	14				

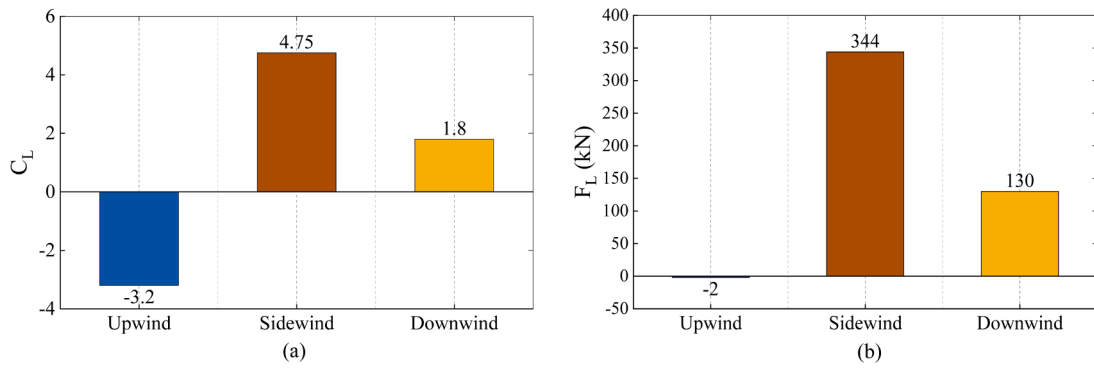


Fig. 5. (a) The lift coefficient and (b) lift in three typical wind directions of a sail.

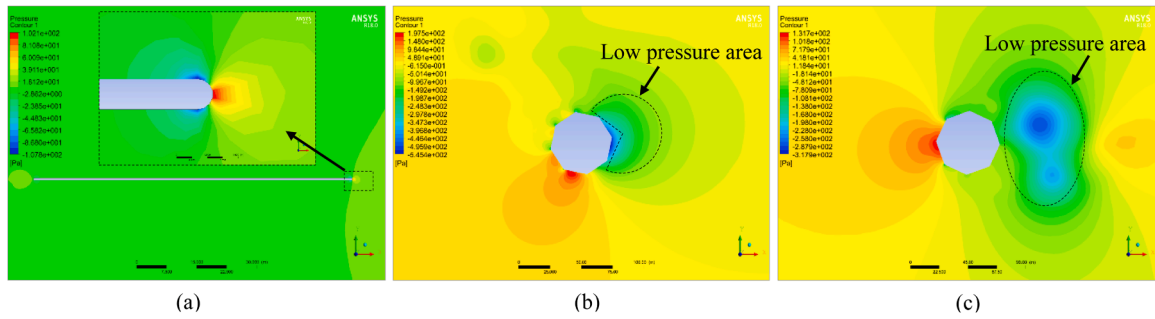


Fig. 6. The pressure distribution of a sail in three kinds of typical wind directions: (a) Upwind, (b) Sidewind, (c) Downwind.

downwind, the shape of the sail is a static polygon with a lift coefficient of 1.8, which can generate a lift of 130 kN. When the device is subjected to sidewind, the polygonal sail is rotating and the device generates Magnus lift. From the histogram, it can be seen that the lift coefficient of the sail is 4.75 and the lift is 344 kN, which means that the thrust generated by a rotating sail is greater than that of a stationary sail.

Fig. 6(a)–(c) are the pressure distribution around the sails under the conditions of upwind, sidewind and downwind, respectively (The figure shows the x-y plane, $z = 7$ m, and the bow direction is x). The sail is closed in the upwind condition, it can be seen from Fig. 6(a) that the pressure is mainly concentrated on the windward side of the sail, the force is resistance for sail. The sail is a polygon in sidewind condition, it can be seen from Fig. 6(b) that the Magnus effect is generated due to the effect of the sidewind and the counterclockwise rotation of the sail, and a positive pressure difference is appeared along the direction of the ship's voyage, so it is said that the new type of rotating sails can generate

thrust for the navigation of vessels. Fig. 6(c) is the pressure distribution around the sail when the vessel downwind, when the wind flows in the same direction as the movement of the ship, polygonal sails play the role of a traditional sail. Therefore, the sail can also provide the vessel with thrust contribution in a downwind condition.

4.2. The comparison and analysis for nine groups of sails

4.2.1. The lift coefficient and drag coefficient corresponding to the polygonal sail

The above section 4.1 explains that when the vessel is subjected to sidewind, the rotating octagonal sail can produce Magnus force, that is, the polygonal sail can provide thrust to the vessel. The purpose is to compare the aerodynamic effects of polygonal sails, the lift and drag effects produced by polygonal sails in sidewind are given in the following three spin ratios ($k = 1, 2, 3$).

Fig. 7 is a point line diagram of the lift coefficient and drag coefficient of nine groups of polygonal sails when the spin ratio k is equal to 1, 2, and 3 respectively. Observing the lift coefficients in Figs. 7(a), 7(c), and 7(e), it can be found that the value range of the lift coefficient is 2.44–8.30. It can be seen from the figure that under the conditions of 5 kinds of wind levels, the lift coefficient shows an overall upward trend with the increase of the number of sides of the sail. This upward trend is more obvious when $k = 1$ (Fig. 7(a)); $k = 2$ (Fig. 7(c)), the lift coefficients of the three kind of sails with sides 16, 18, and 20 are basically the same. When $k = 3$ (Fig. 7(e)), the lift coefficients of the three kind of sails with the number of sides 16, 18, and 20 are basically equal to those of the n -sided sail. This shows that the installation of 16-sided sails on the vessel, the propulsion effect can be close to the effect of circular sail (n -sided) of equal diameter when $k = 3$. In addition, as the level of wind increases, the lift coefficient decreases, but this does not mean that the lift produced by the sail becomes smaller.

Observing Figs. 7(b), 7(d), and 7(f), it can be seen that the greater the wind level, the greater the drag coefficient of the sail, which is the opposite of the above-mentioned lift coefficient trend. Fig. 7(b) is a graph of the drag coefficient at $k = 1$. It can be seen from the graph that as the number of sides increases, the drag coefficient shows a downward trend as a whole. Fig. 7(d) is the drag coefficient graph at $k = 2$, the drag coefficient with different numbers of sides has a smaller change of steepness. Observe Fig. 7(f), when the spin ratio $k = 3$, the drag coefficient is larger overall, and its range is mainly 1.5–1.7.

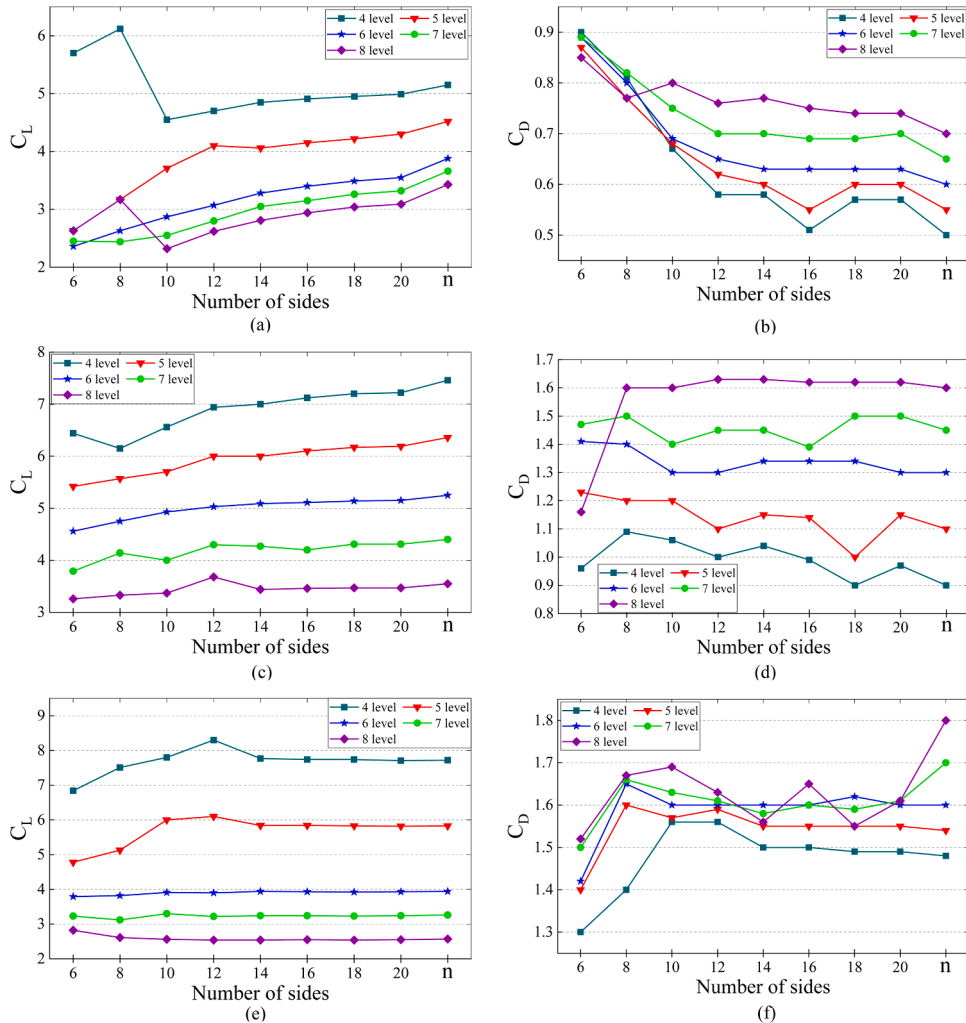


Fig. 7. The lift coefficient and drag coefficient corresponding to the polygonal sail: (a, b) $k = 1$, (c, d) $k = 2$, (e, f) $k = 3$.

4.2.2. The pressure distribution around the polygonal sail

The following gives the pressure distribution around nine groups of polygonal sails when the wind level is 6 and the spin ratio is 2. Among them, the pressure distribution uses the sail cross section (x - y plane) in $z = 7$ m, as shown in Fig. 8.

Fig. 8 is the pressure distribution around the polygonal sail under the condition of $k = 2$ and 6-level wind. Observing the picture, it can be seen that due to the sail rotates in a counterclockwise direction, a low pressure area is generated on its right side; as the number of sides of the polygonal sail increases, the difference value of pressure in the x -direction generated around the sail becomes larger and larger; after the number of sides of the sail reaches 16, the variation of pressure difference for the three graphs in Figs. 8 (f-h) is no longer obvious. By comparing the pressure of different polygonal sails, it can be proved that the number of sides is one of the main factors that affect the Magnus effect of the sail. Due to the different of pressure values in the x -direction (along bow direction), the sail can contribute to the vessel's thrust. In addition, it can be seen from the figure that the pressure gradients are discontinuous on the high pressure side of the polygonal sail.

Besides, in order to make a more intuitive comparison, the pressure on both sides of polygonal sails with 6, 10, 14, 16, 18 and n edges along the x -direction are measured. To be special, dividing the pressure area into two parts with $x = 0$ as the dividing line, and pressure curves graph are made, as shown in the Fig. 9, the position in the y -direction has displayed in the x -axis; the y -axis is the pressure value. For example, the pressure curve in Fig. 9(a) represents the position of the marker in Fig. 8

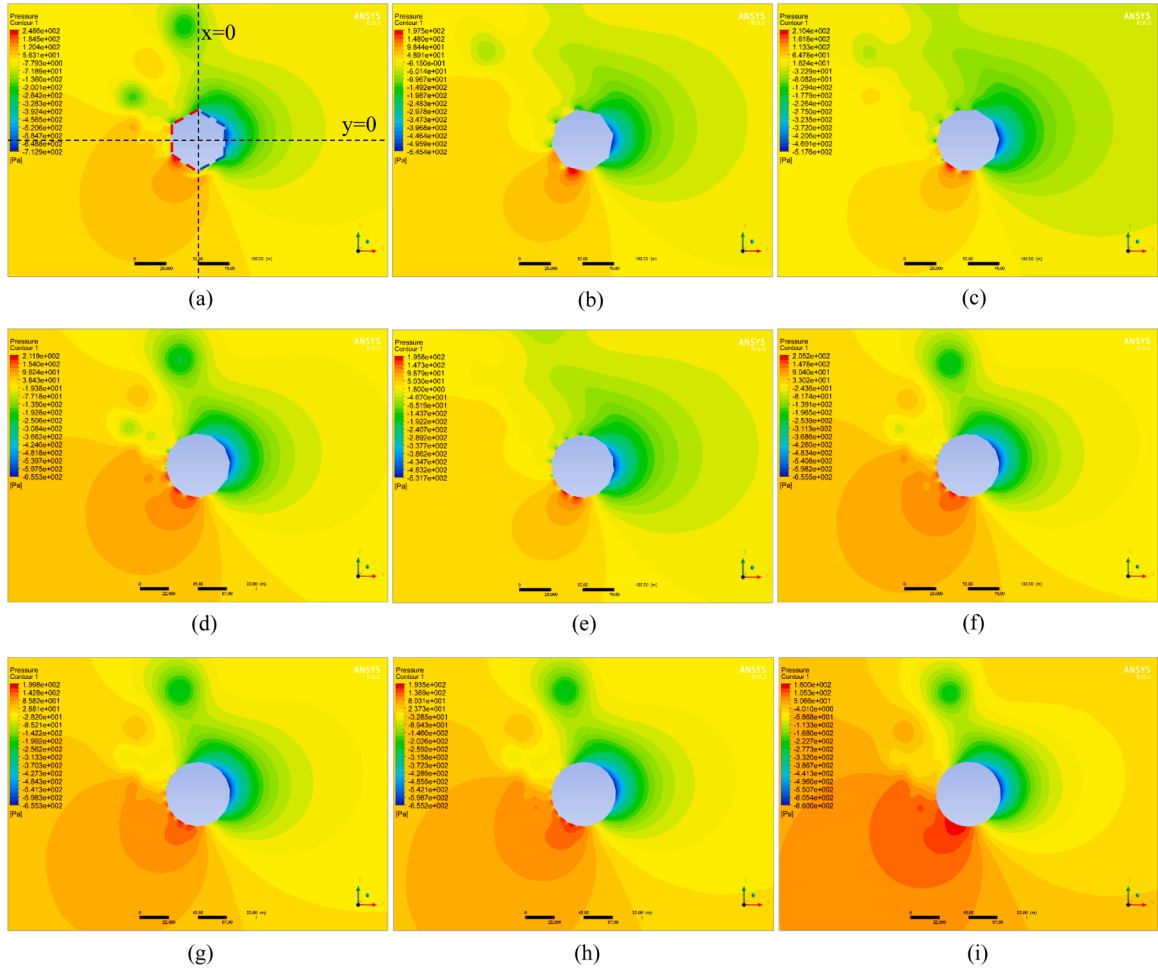


Fig. 8. Pressure distribution around the polygonal sail: (a) Hexagonal Prism, (b) Octagonal Prism, (c) Decagonal Prism, (d) 12-Sided Prism, (e) 14-Sided Prism, (f) 16-Sided Prism, (g) 18-Sided Prism, (h) 20-Sided Prism, (i) Circular Cylinder.

(a).

On the whole, the high pressure side curve of the polygonal sail is shown in Figs. 9(a-e) is not smooth. However, from the pressure distribution in Fig. 8(i) and the red curve in Fig. 9(f), it can be seen that the pressure distribution of the circular sail is relatively uniform. Observing Fig. 9(a), at the position $y = 0$, the pressure difference on both sides of the hexagonal sail is 640 Pa. It can be seen in Fig. 9(b) that near $y = 0$, the pressure difference on both sides of the 10-sided sail is 520 Pa. Similarly, in Fig. 9(c), the pressure difference of the 14-sided sail is 550 Pa. Fig. 9(d) is the pressure curve of the 16-sided sail. It can be seen from the figure that at the position of $y = 0$, the pressure difference of the 16-sided sail is about 750 Pa. In Fig. 9(e), the pressure difference of the 18-sided polygon is about 750 Pa. Fig. 9(f) is a circular pressure curve, its two lines are relatively smooth. The pressure difference near $y = 0$ is 700 Pa. All of these pressure differences are positive. To a certain extent, the pressure distribution of middle section ($z = 7$) can represent the trend of the pressure distribution around the whole sail. These pressure difference values can explain: the values of the pressure difference along the vessel's sailing direction are positive. Looking closely at Fig. 9, it can be seen that as the number of sail sides increases, the area formed by the red and blue lines in the figure becomes larger and larger, the pressure difference become increase, this phenomenon is the same as the changing trend of the pressure distribution shown in Fig. 8.

By comparing the lift coefficient curves and pressure distribution in Fig. 7, Fig. 8, and Fig. 9, it is found that after the number of sides of a polygonal sail reaches 16, as the number of sides of the sail increases, its lift coefficient is basically unchanged; as the number of sides of the sail

increases, the change of the pressure difference value along the x -direction is no longer obvious. In addition, as the number of sides of the sail increases, that is, the number of cylinders in the device increases, the mass of the device and the complexity of the changing shape also increase. Therefore, considering the simulation results and other factors comprehensively, 16-sided sails are selected as the optimal value here, that is to say, the new assisted propulsion device has 16 cylinders. The following mainly focuses on the 16-sided sail for further research.

4.3. Thrust contribution of polygonal sails

To assess the potential of polygonal sails as a device for the vessel propulsion, for the sail on a vessel, it is important to consider that the resulting wind speed is the vector sum of the vessel speed and the environmental wind. So, wind angle (θ) is must be considered in practical application, the thrust (F_T) acting on the vessel is determined by both the lift and drag of the polygonal sail, referencing the equation (8).

$$F_T = F_L * \sin\theta - F_D * \cos\theta \quad (8)$$

The vessel is encountered wind of different direction in the ocean environment, Fig. 10 shows the values of thrust coefficient (C_T) and thrust (F_T). Among, for comparative analysis, the speed of the vessel has been fixed at 10 knots. When the wind angle is 0° , the device is upwind condition as shown in Fig. 1(c). When wind angle is 0° and 180° , the polygonal sail does not rotate, others do rotate. The thrust coefficients of the sails corresponding to the three spin ratios ($k = 1, 2, 3$) are given in Fig. 10(a) when the wind speed is 13 m/s. When the spin ratio is 2, the

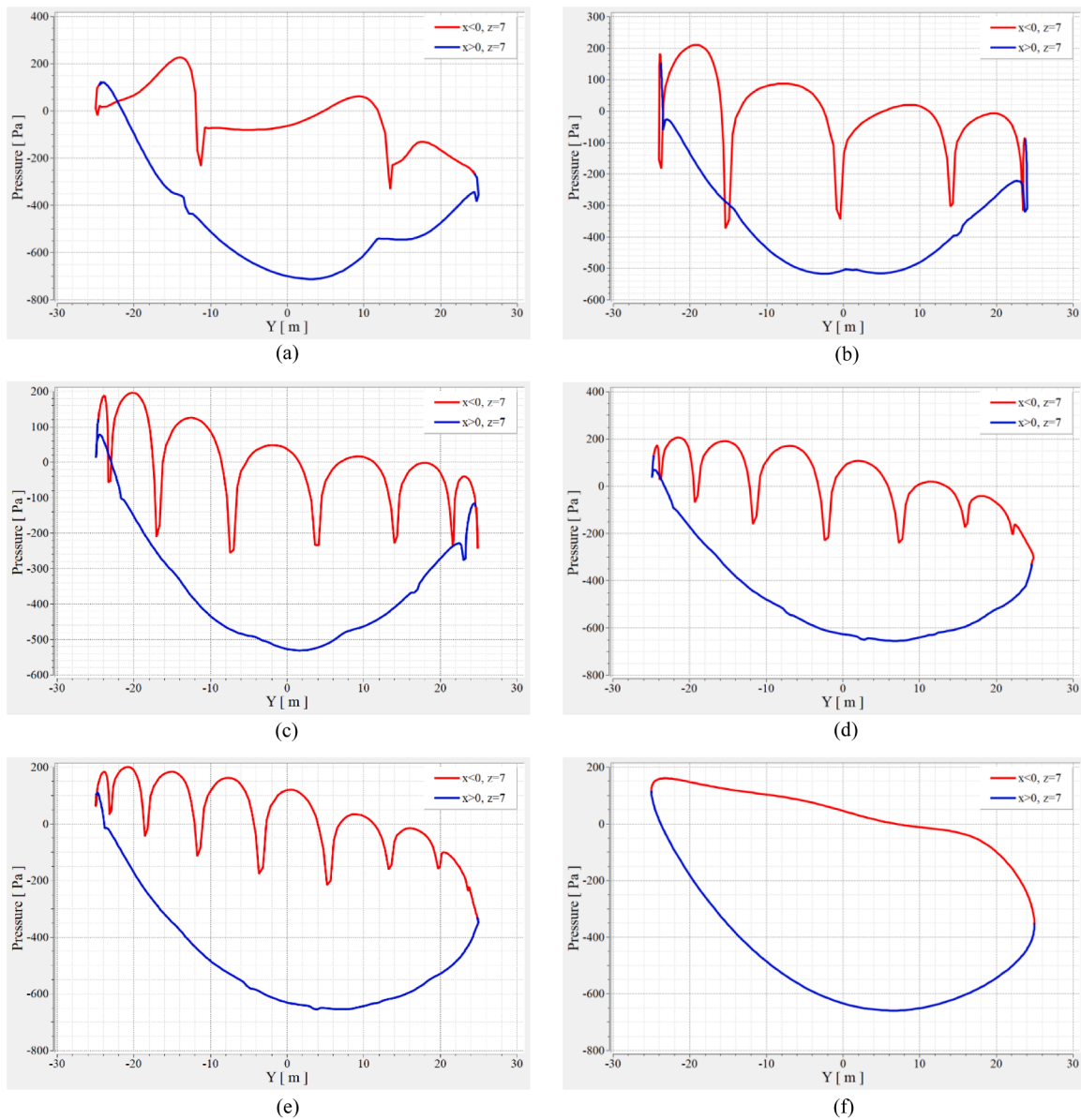


Fig. 9. Pressure curves of part of polygonal sails: (a) Hexagonal Prism, (b) Decagonal Prism, (c) 14-Sided Prism, (d) 16-Sided Prism, (e) 18-Sided Prism, (f) Circular Cylinder.

thrust corresponding to the five kinds of winds is given and it is shown in Fig. 10(b).

Fig. 10(a) shows that thrust coefficients of sail for a variety of wind angles. The wind angle range is 0-180 degrees and 180-360 degrees corresponding to the same response of the sail. As the wind angle changes from 22.5° to 157.5° , the lift coefficients of the three spin ratios gradually increase, maximum lift coefficient is 12, it appears at $k = 3$ and 157.5° . A closer look at the zero points of the thrust coefficient shows that different spin ratios correspond to different wind angles at the zero points, roughly, they are 22.5° , 45° and 67.5° respectively at $k = 1, 2, 3$. Besides, when the vessel is sailing downwind, the thrust coefficient of the sails is approximately 0.6.

Fig. 10(b) shows the sail thrusts corresponding to a variety of wind angles. When the wind angle is less than 45° , the sail has a negative effect because the lift component of the sail is less than the drags for small wind angles in the sailing directions. When the wind angle is greater than 67.5° , the thrust tends to increase and then decrease as the wind angle increases, the maximum thrust is 540 kN at 112.5° and wind of 8 level.

4.4. Power generated by the new assisted propulsion device

It is important to study the actual energy contribution that the sail brings to the vessel. The power consumed by the motor that rotates the sail (P_{motor}) and the output power of the sail (P_L) determine the power of the main engine that the rotating sail can replace. Therefore, the propulsive power generated by the 16-sided sail for 5 kinds of wind conditions is given in Fig. 11. In addition, in order to have better analysis and comparison of the propulsive power, the vessel's speed (V_{ve}) is still used here at 10 knots.

As in Fig. 11, the three bar graphs are the output power (P_L) and propulsive power ($P_{prop.}$) of the corresponding 16-sided sails for $k = 1, 2$, and 3 conditions, respectively. On the whole, when the spin ratio k is 1, 2 and 3 respectively, it is an increasing trend of output power of the sail as the wind level increases, and the maximum output power appears when the wind level is 8, which are respectively 2519 kW, 2964 kW and 2182 kW.

Since the rotor's energy consumption of the propulsive power is considered, a more valuable information compared to the output power

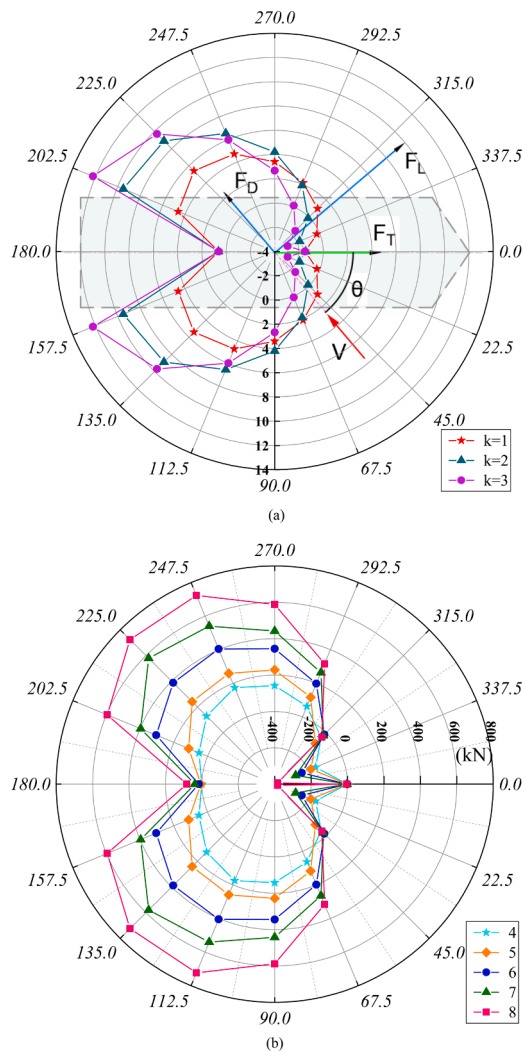


Fig. 10. the (a) is the C_T for a defined sail of 16-sided at different values of k and wind angle. (b) the F_T delivered by the sail for the different magnitude of relative wind levels (fixed $k = 2$).

is the propulsive power. Combined with the analysis in Fig. 11 and Table 6, when k is 1, the maximum value of propulsive power appears at the 8-level wind, it is 2005 kW, it is also the maximum propulsion contribution provided by the polygon sail to the vessel in 4-8 level wind conditions. The maximum value of the propulsive power appears at the 8-level wind at $k = 2$, it is 1935 kW. When $k = 3$, the maximum value of the propulsive power appears at the 5-level wind, its value is 1059 kW. Observing Table 6, the power consumed by the motor accounts for 15%-35% of the output power when the maximum propulsive power appears. This indicates that the propulsion contribution made by the new assisted propulsion device for the sailing vessel is much larger than the consumption caused by its own rotation.

4.5. Effect of Reynolds number on the new assisted propulsion device

In experiments and simulations on rotating cylinders, many scholars have suggested that Reynolds number is one of the main factors affecting the lift and drag coefficients of the cylinder [36,37], and the following explores the effect of Reynolds number on the new assisted propulsion device. Specifically, the lift-to-drag ratio of the 16-sided sail is calculated for five kinds of Reynolds number conditions, i.e. (C_L/C_D). Where k is the spin ratio; the air viscosity (ν_a) is 1.45×10^{-5} Pa/s.

Fig. 12 shows the lift-to-drag ratios of polygon sails at different

Reynolds numbers. Look at the figure, it is a decreasing trend of the lift-to-drag ratio as the Reynolds number increases. When Reynolds is same, the lift-to-drag ratio decreases sequentially with spin ratio k is from 1 to 3. The maximum lift-to-drag ratio presented in the figure occurs at $Re = 2.41 \times 10^7$ and $k = 1$, and its value is 9.63. In addition, the steepness of the dotted line diagram is different for different spin ratios, but their trends are the same, and the steepness decreases when the Reynolds number over 4.48×10^7 . In summary, like the general circular sails, the Reynolds number is also the main factor affecting the force coefficient of polygon sails.

5. Conclusions

This research proposes a new assisted propulsion device of the vessel, which is a polygonal sail composed of a sail and multiple cylinders. The sail is essentially a large Flettner rotor with a large projection area. In addition, when the vessel sails is upwind conditions, the device can automatically change its shape, and the device can effectively reduce the wind resistance. The main structure of the device is modeled using SolidWorks, a three-dimensional computational domain containing polygonal sails is established. Polygonal sails are simulated and calculated using Fluent software under various conditions, and further analyze the propulsion contribution that sails can make to vessels. The specific analysis results are as follows:

- 1 After the sail is simulated under the three typical conditions of upwind, sidewind and downwind, it was found that the resistance generated by the sail itself was very small during vessel is sailing against the wind, the resistance about 2 kN. Under the sidewind condition, due to the Magnus effect produced by the rotation of the sail, the device exhibits a good propulsion effect.
- 2 A comparative analysis of the lift coefficient and drag coefficient of nine groups of polygonal sails (including circular sails), it is found that the growing trend of the lift coefficient as the number of sail sides is increasing. Moreover, when the number of sides of a polygonal sail reaches sixteen, increasing the number of sides again, the lift coefficient is continual augment, but the rate of increase is very less, so, the lift coefficient of the 16-sided sail can achieve a good lift effect compared to other polygonal sails.
- 3 From the pressure distribution in Fig. 8 and the curves in Fig. 9, due to the difference value of pressure in the x-direction (the bow direction), the sail can make contribution to the vessel's thrust. In addition, the pressure gradients are discontinuous on the high pressure side of the polygonal sail.
- 4 Under certain lateral wind conditions, when the wind angle increases, the thrust coefficient gradually increases, with the maximum thrust coefficient reaching 12 at $k = 3$. When the wind angle is 112.5° , the corresponding thrust of the sail reaches its maximum, the maximum value is 540 kN in 8-level wind. In addition, when the wind angle is equal, the polygonal sail generates an increasingly large value of thrust as the wind level increases.
- 5 Under $k = 2$ and 8-level wind conditions, the output power of the 16-sided sail reaches 2964 kW. Considering the power consumed by the rotation of the sail, the maximum propulsive power produced by the sail is 2005 kW at $k = 1$ and 8-level wind. In addition, according to Table 6, when the maximum propulsive power appears, the power consumed by the assisted propulsion device accounts for 15%-35% of the output power. This indicates that the device's contribution to the propulsion of sailing vessels is far greater than the loss caused by its own rotation.
- 6 It is similar to a general circular sail, the Reynolds number is also one of the main factors affecting the force coefficient of a polygonal sail. As the Reynolds number increases, the lift-to-drag ratio of the sail becomes smaller.

A new assisted propulsion device is proposed, and its main structure

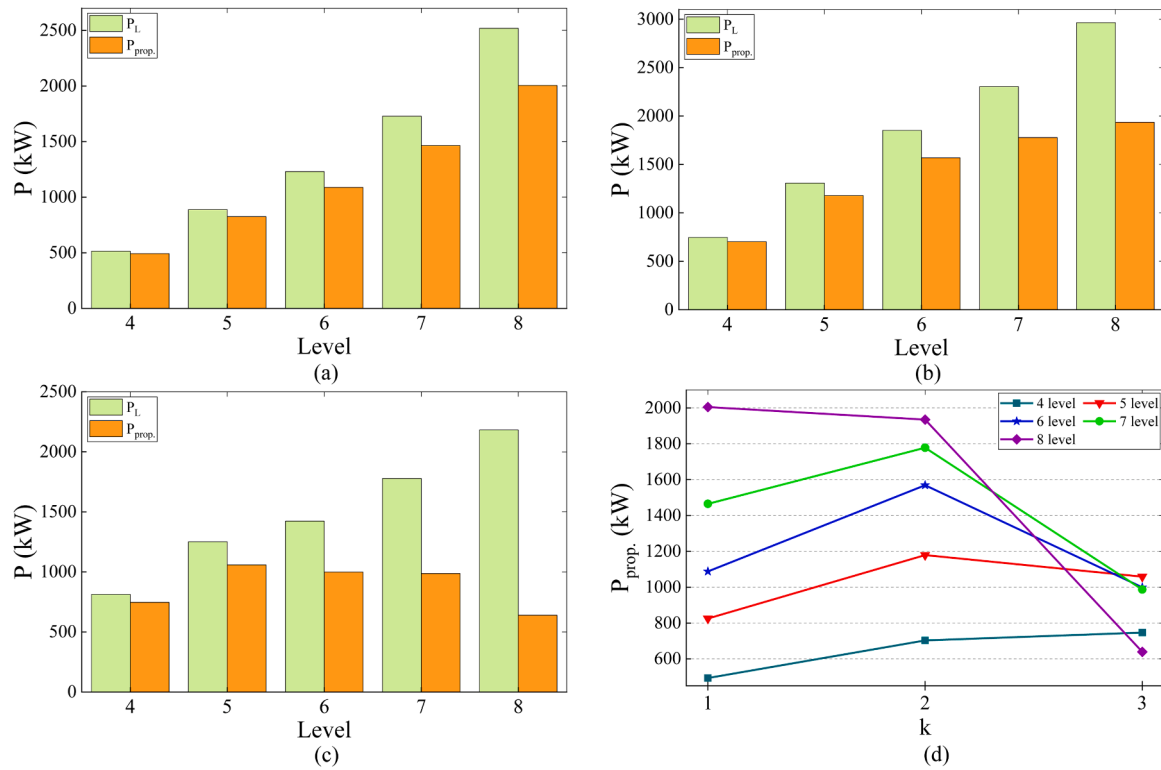


Fig. 11. Propulsive power of 16-sided sail: (a) $k = 1$, (b) $k = 2$, (c) $k = 3$, (d) Summary of propulsive power.

Table 6

Comparison of power at maximum propulsive power time.

k	Maximum Propulsive power (kW)	Corresponding wind level	Corresponding output power (kW)	Corresponding consumed power (kW)	Consumed/Output power
1	2005	8	2519	514	20.4%
2	1935	8	2964	1029	34.7%
3	1059	5	1251	192	15.4%

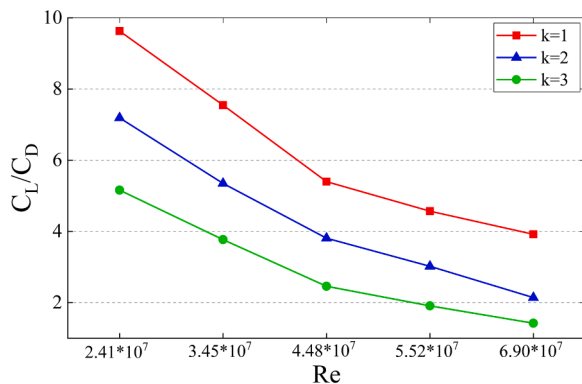


Fig. 12. Lift-to-drag ratio of the polygonal sail under different Reynolds number conditions.

and use method are briefly studied. The dimensions of the device, including diameter and height, were modeled for a 300000-ton tanker, and a 16-sided sail was found to be of good practical value. However, for other medium to large merchant vessels, the device can also be modified and installed on the vessel, for example, if the device is installed on the deck of a smaller merchant vessel, the diameter of the device, the height, and the number of the sail sides can be made smaller. Therefore, this study not only proposes a polygonal sail to provide propulsion to the vessel, but also proposes a new idea to optimize the Flettner rotor. At

present, although this research is in the stage of simulation experiment, and some structures and calculations are idealized, reasonable simulation experiments can pave the way for later engineering experiment applications.

Authors' Contributions

- (1) Boyang Li: Experimental design, Writing- Original draft preparation
- (2) Rui Zhang: Simulation calculation, Writing- Reviewing
- (3) Baoshou Zhang: Visualization (Pictures and tables.)
- (4) Qianqian Yang: Data processing
- (5) Chao Guo: Investigation of Sea Conditions

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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